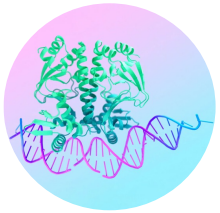
SciLifeLab

AI agents in action



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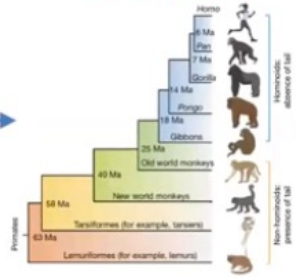
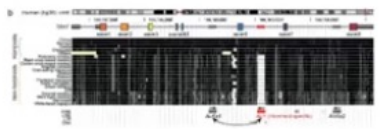
Life science workflows are complex, diverse, and abundant

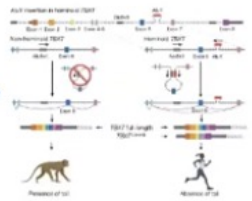
Geneticist <> Bioinformatician <> Wet-lab technician <> …

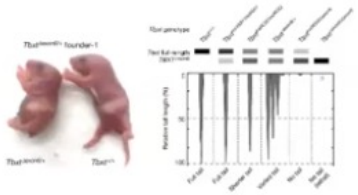
Q: what is the mechanism of human losing the tail?

Database → Bioinformatics → Reasoning → Wet-lab









Reasoning → Experiments & analysis (action)

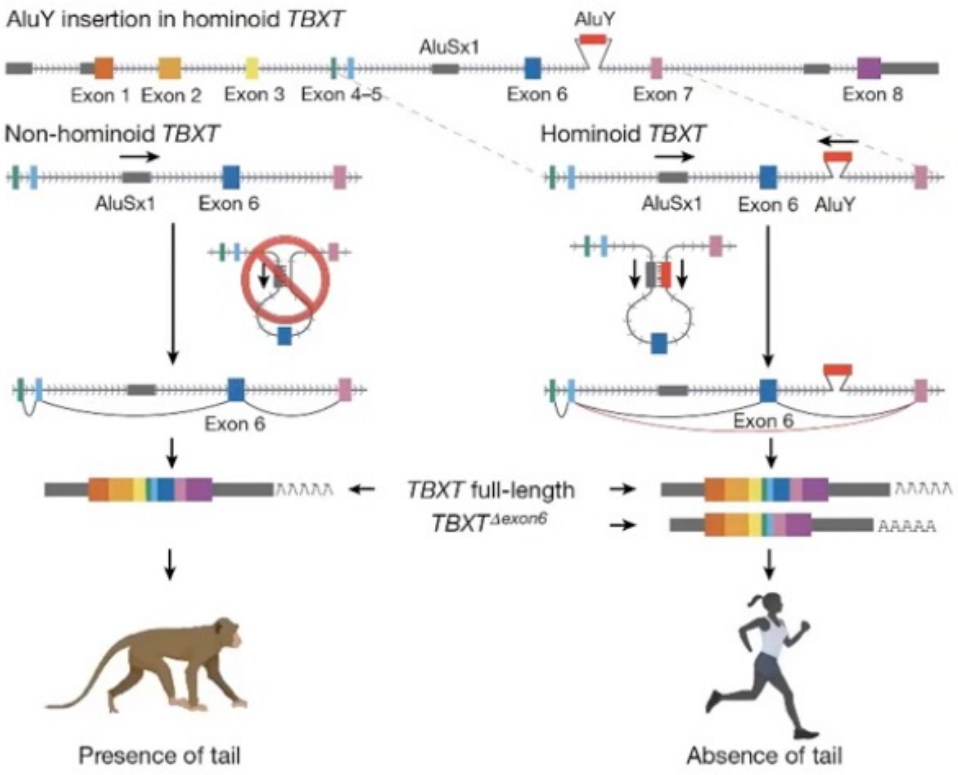
It is intrinsically a loop of reasoning + actions!

Biomni talk: https://www.youtube.com/watch?v=kOIrpy8V7GM f]

How is Life science research being done today?

(1) Generating a hypothesis

"TBXT causes tail loss."



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How is Life science research being done today?

(2) Validating a hypothesis

Biostatistician

Reasoning why a variant could cause a phenotype.

Statistical analysis from massive data generates p-values

GTEx Portal

RCSB PDB PROTEIN DATA BANK

biobank uk — Enabling scientific discoveries that improve human health

GWAS Catalog

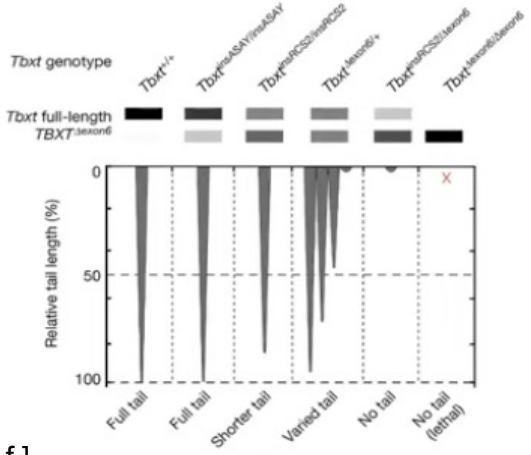
CZ CELLxGENE

ENCODE

biologist

A high throughput screen to search for genes that regulate a phenotype.

Targeted wet-lab experiment



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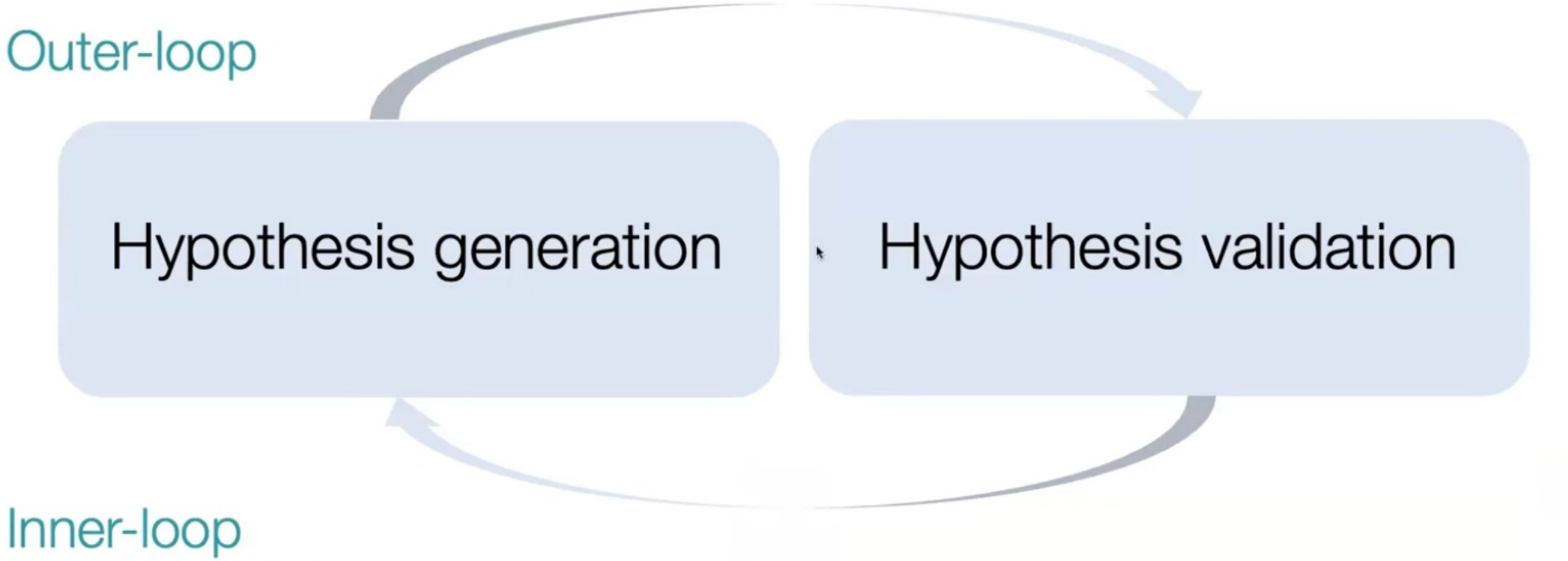
Life science research at a glance



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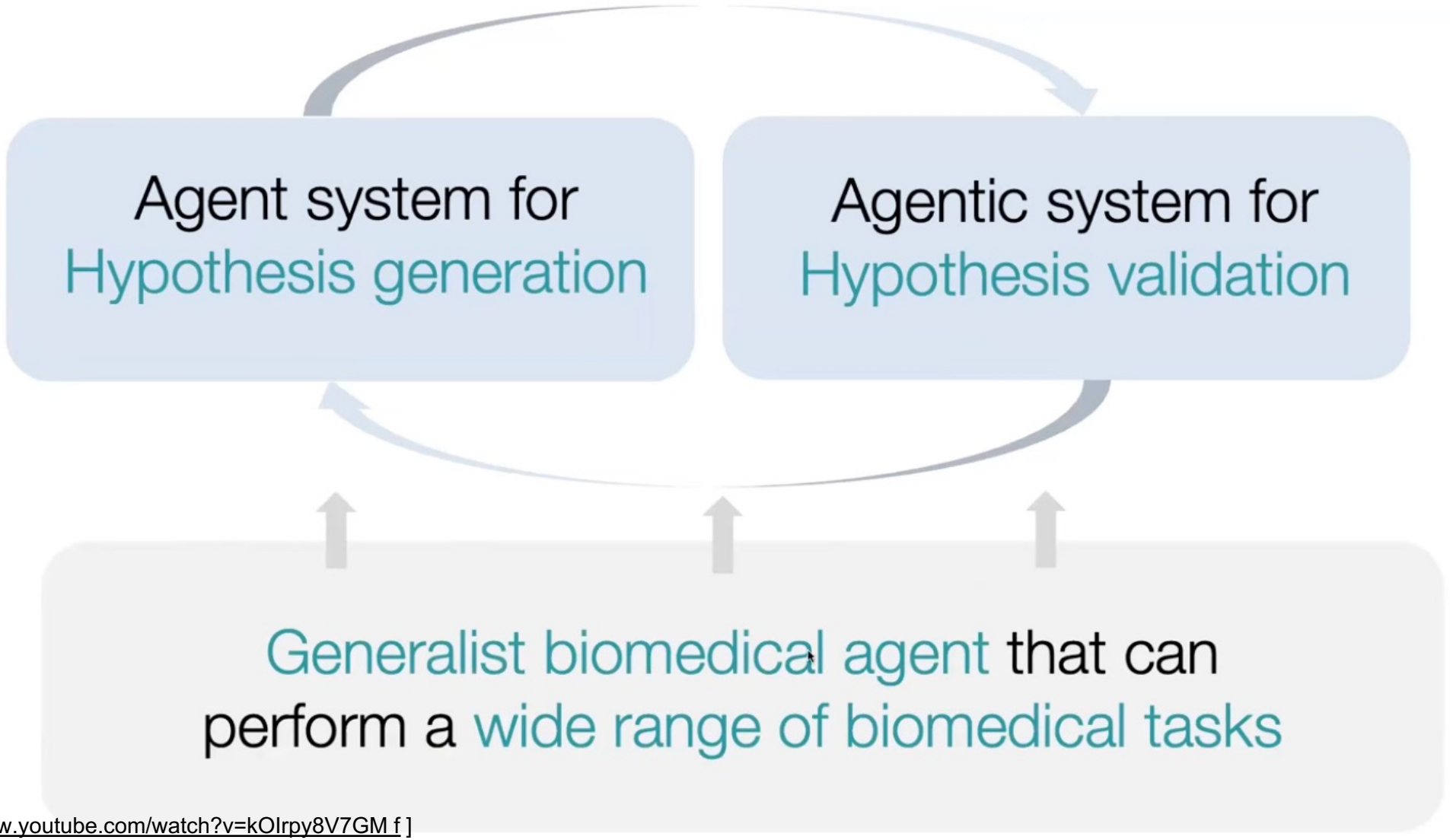
A framework for building collaborative AI agents in life science



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In the quest for a life science agent…

Capability	Details
Adapt	Plug in tools from genomics, drug discovery, clinical data, … any domain
Unify	Orchestrate workflows across multiple domains
Evolve	Add new skills as research advances
Efficient	Automate repetitive data tasks
Reproducible	Transparent, executable record of every step
Scalable	Tackle complex, multi-step research at scale

Finding the right agent for the job

If you need to...	Consider...
Analyze RNA-seq/scRNA-seq data	AutoBA, CellAgent
Explore spatial transcriptomics	SpatialAgent
Understand gene functions	GeneAgent
Design CRISPR experiments	BioDiscoveryAgent
Research drug interactions	TxAgent
Query biomedical knowledge graphs	BioChatter, ESCARGOT
General biomedical research tasks	BioChatter, Biomni
Clinical/medical decision making	MDAgents

Emerging trends

Trend	Examples	Impact
Self-verification	GeneAgent, ESCARGOT	Reduces hallucinations in scientific outputs
Multi-agent collaboration	MDAgents BioChatter	Simulates expert team consultations
Tool-augmented reasoning	TxAgent Biomni	Grounds AI in validated databases
Code generation + execution	AutoBA, CellAgent, Biomni	Enables end-to-end analysis pipelines
Knowledge graph integration	ESCARGOT, BioChatter	Structures biomedical relationships

Thank you! 😊

Contact: datacentre@scilifelab.se, serve@scilifelab.se

Explore the AI agent landscape tool 😊

Use the link below, or scan this QR code

https://shorturl.at/MOY7d

The future belongs to those who adapt! 😊